UNIVERSITY OF DAR ES SALAAM
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY
DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING

Name: Musyani, Samuel Dan | Reg. Number: 2022-04-09050
Supervisor: Dr. Hashimu Iddi

TITLE: DESIGN AND IMPLEMENTATION OF AN AUTOMATIC ID CARD ISSUANCE SYSTEM

PROBLEM

The current UDSM student ID-card collection process is manual. Students wait for cards to reach their colleges, then queue during office hours while staff search and sort large batches.

≈ 1 week	Long queues	Office hours
before cards reach some colleges	after cards are released	limit student access

OBJECTIVES

Main objective

Design and implement a secure self-service system that dispenses pre-printed student ID cards on demand and records every transaction.

- Detect each card and extract the registration number using OCR.
- Authenticate students using registration number, SMS OTP and PIN.
- Select the correct indexed slot and dispense only one card.
- Maintain audit logs and evaluate retrieval, authentication and error handling.

PROPOSED SOLUTION

A two-tier automated kiosk combines high-level computing with real-time mechanical control.

- Raspberry Pi 5: touchscreen interface, OCR, student verification, OTP/PIN authentication, database access and logging.
- Arduino Mega 2560: receives SPI commands and controls servo-driven card positioning and ejection.
- Indexed carousel storage links each verified student record to a physical card slot.

EXPECTED IMPACT

24/7	Fewer	Lower	Audit-able
student access	queues	staff workload	transactions

HOW THE SYSTEM WORKS

STAFF PRE-LOAD AND NOTIFICATION

1 Load cards	2 Capture + OCR	3 Validate + map	4 Notify
Staff places pre-printed cards in the kiosk.	Camera reads each registration number.	Student record and storage slot are assigned.	OTP and temporary PIN are sent by SMS.

STUDENT RETRIEVAL

5 Identify	6 Authorize	7 Dispense	8 Log
Student enters registration number, OTP and PIN.	Credentials and card availability are verified.	SPI command activates the selected card slot.	Collection status and audit record are updated.

SYSTEM ARCHITECTURE

INPUTS
Camera / OCR
UI Screen
SMS gateway
Student database

›

RASPBERRY PI 5
High-level controller
UI • OCR • Authentication
Database • Notifications
Audit logging

›

ARDUINO MEGA 2560
Real-time controller
Servo control
Slot positioning
Card ejection

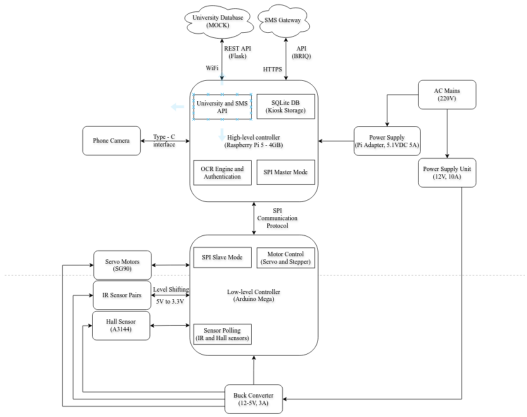
PROTOTYPE STATUS

- Functional OCR card detection and institutional registration-format validation were achieved together with audit logging of all activities in the system.
- SMS OTP, PIN verification, secure credential hashing and three-attempt lockout were implemented.
- SPI messaging and servo-control tests validated communication between the Raspberry Pi and Arduino Mega.

CONCLUSION

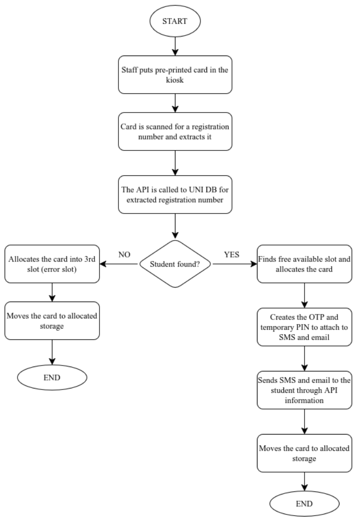
The prototype confirms that a secure, auditable and scalable self-service kiosk can automate the distribution of pre-printed student ID cards.

SYSTEM BLOCK DIAGRAM

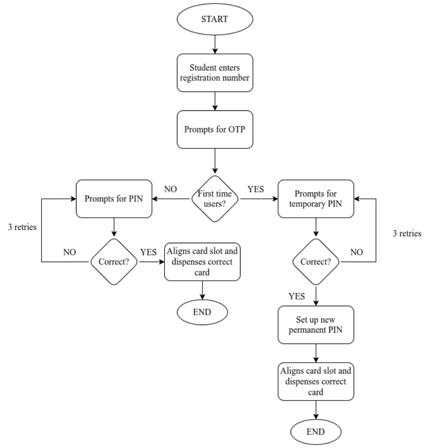


FUNCTIONAL FLOWCHARTS

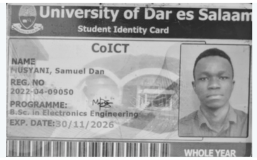
STAFF SIDE

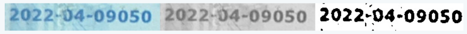


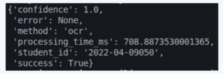
STUDENT SIDE



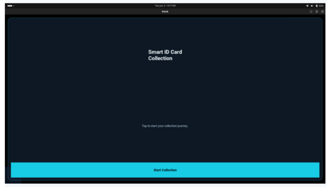
COMPUTER VISION RESULTS

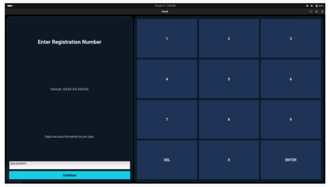


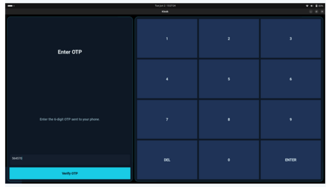


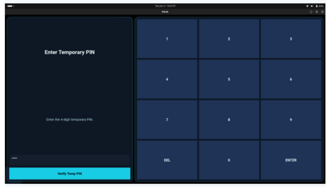


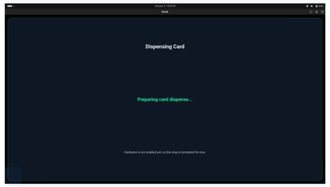
CRUCIAL USER INTERFACE PAGES

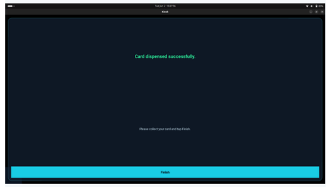












FINAL PROTOTYPE RESULT